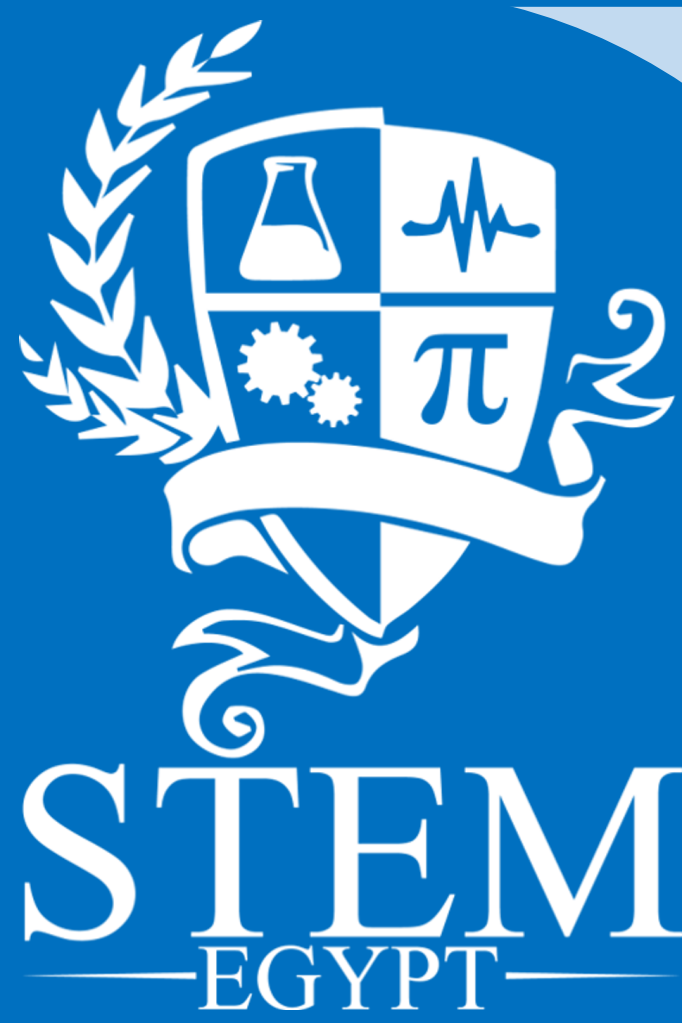




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SWING LINK

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ABSTRACT

Egypt faces some challenges that threaten its economy, hindering its development. Among those challenges are urban congestion and overpopulation. Overpopulation is the increase in the people number that live in a certain region which leads to the urban congestion problem. Urban congestion is the excessive accumulation of people and activities in urban areas. The purpose of the study is to solve the urban congestion problem in El-Qalyoubia governorate, especially in Kafr El-Shorafa El-Gharby and Bahada villages. The problem is that there is high population in the two villages which make the move between them too hard and the deficiency in some services in one of the villages that is more available in the second village. The bridge is a swing bridge. It is provided with a central pier rotating. The bridge moves around itself around a central pier. When normal traffic has to cross the bridge, it rotates to its original position as a fixed bridge to let the traffic cross. When a ship has to pass the bridge, the bridge rotates at a 90-degree angle from its fixed position to allow ships to cross. The Pratt truss is used in the bridge because it can distribute all the vertical loads effectively. The bridge met the design requirements as the bridge's mass didn't exceed 20 kg mass, it lifted a 5kg load on it without collapse. The bridge made a deflection 1.8mm which is less than the requirement.

INTRODUCTION

Egypt's grand challenges are obstacles that hinder Egypt's development and weaken its economy, agriculture, and industry. Population growth is one of Egypt's grand challenges. It's the increase in population in an area that makes it difficult for the citizens to live. In 2024, it reached 116,538,258 people, a 1.75% increase from 2023. population growth causes Urban congestion, which is the excessive accumulation of people in urban areas, which increases population density. As shown in Figure 1, population density increases in the governorates near the Nile River. So, the specific problem to be solved is urban congestion due to the inability to cross water streams in El-Qanatir El-Khairia region near the Nile River, which increases traffic jams and greenhouse gas emissions. Solving this problem will decrease the urban congestion in those areas.

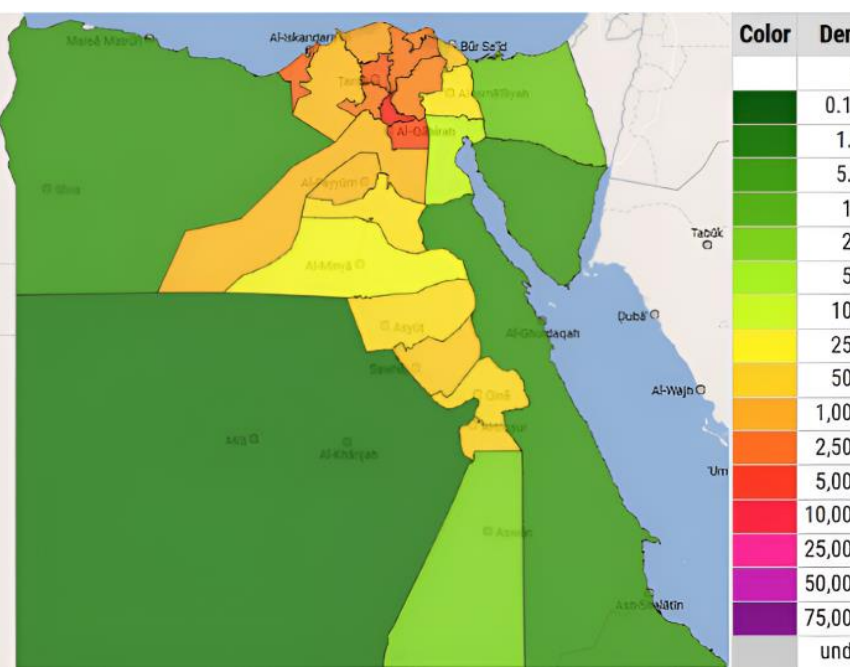


Figure 1: Population density in Egypt's governorates

One of the prior solutions is Imbaba Bridge, which was built in Cairo in 1925. The bridge has strong wind durability because of its weight. The only con of the bridge is that it takes a long time to operate because of the complex and heavy mechanical parts. Another prior solution is Burnside Bridge. The bridge was built in 1926 in Portland, USA. The bridge requires less amount of energy to be operated, but it has many problems because of its old age such as the need to get maintenance at a high cost.

The selected solution is to make a swing bridge with a center pivot and the movable part rotates around it. The truss is a Pratt truss that distributes the vertical forces applied on the bridge effectively with its simple design. It needs to meet the design requirements of 80 to 100 cm length, 22cm width, contains two traffic lanes for vehicles of 7cm width each and two pedestrian lanes of 3cm width each. Also, the prototype's mass shouldn't exceed 20kg and can carry about 5kg mass on it, with a maximum deflection of 5cm without collapsing. The clearance between the lowest point of the bridge and the water level will be 10 centimeters to allow boats with less than 10cm height to pass through easily. The mechanism should allow ships to cross the water stream 28 centimeters wide.

MATERIALS & METHODS

Seven main materials were used in making the prototype, as shown in Table 1:

Table 1: materials used to make the prototype.

Name	Description	Quantity	Picture
Palm tree midribs	The central structural part of a palm leaf, used in truss and deck structures.	35 meters	
Gears	Two big compound gears (42 teeth) hold the movable deck and one small gear (18 teeth) for the transmission of Torque and rotational motion from the manual lever to the bridge's rotating mechanism, they also support loads.	3 gears	
Chain	Links both gears together to move the motion between them.	1 chain (147 cm)	
MDF spray	A strong adhesive spray provides fast drying and an even adhesive layer. It was used to connect truss members and connect palm midribs to form the deck.	2 cans	
Metallic wire	A metallic wire was used to link the larger gear with the bottom of the deck.	0.2 meters	
Ball bearing	Tow metallic ball bearings are put inside the gears to reduce friction and improve their efficiency.	2 ball bearings	
White glue	A strong adhesive material that takes a long time to dry but sticks strongly	1 pack (250 gm)	

1- The dimensions were selected, and a basic 3D model was made using SketchUp program, as shown in Figure 2.

2- As shown in figures 3 and 4, The palm midribs were brought and cut into pieces, after measuring them with a ruler and marking the lengths with a pencil, with thicknesses equal to 5mm and lengths of 10cm, 14.1cm, 7cm, 3cm, and 22cm.



Figure 2: Basic 3D design of the bridge

3-As shown in Figure 5 and 6, The members in the selected model were arranged and stuck together with the selected sticky material (MDF glue) to complete the truss. After that, the pieces of 22cm length were arranged and stuck horizontally together to make the deck on the movable part and the fixed part.

4-Columns were made using the pieces of palm midribs by putting them vertically (every piece is next to another piece in a vertical case). Then, two gears were used, and ball bears were put in every one of them to avoid friction among the mechanical parts and a lever was put in one of the gears.

5-One gear (that has a lever) was put on the land next to the movable part and the other gear on the column of the movable part and nuts to prove the gears well. Then, the gears were linked by a chain that moved the torque between them.

6-The movable part of the bridge was compounded with a truss on the column that is in the center of the span with the gear on it. A base and land for the prototype were made. The mechanical parts were put in their selected position and the lever was rotated to rotate the gears then moving the bridge.

Test plan:

The prototype was tested to see if it satisfies the design requirements.

1-Endurance:

A 5kg load was put on the mid-point of the longest fixed span of the bridge.

2-Bridge's mass:

The prototype's mass was tested by weighing it, as it mustn't exceed 20 kilograms.

3-Mechanism:

The bridge's mechanism was tested to ensure that it meets the design requirements by rotating around the center pivot to provide two paths for the ships to pass through.

4-Deflection:

a mass starting from 1kg to 5kg was put in the middle of the fixed span to measure the deflection of this span.



Figure 4: Cutting the midribs into small members.



Figure 3: Measuring and marking lengths of members.

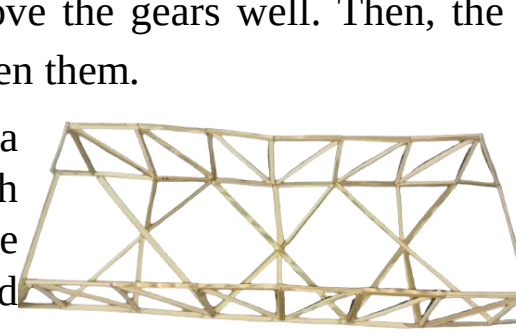


Figure 6: Truss foundation



Figure 5: Sticking the members with MDF spray.



RESULTS

Negative result:

One of the negative results is that the sticky material (MDF) that sticks the members of the deck together didn't bear the weight of 5 kg that was put on it. This is because MDF dries fast but is weak in sticking the members as it forms a thin weak bond. So, The MDF glue was replaced by the white glue as it dries slowly but forms a strong bond between the members which makes the deck more strong and able to carry the weight.

Another negative result was in the testing of the mechanism. When the test was made the mechanism wasn't moving smoothly, instead it was moving with a lot of stops and problems. So, the solution heard was by oiling the chain that links the gears in the system together.

positive results:

A Positive result based on the test plan is that the mechanism of the bridge worked successfully. After that, The 5kg mass test was made by adding a mass of 5 kg to the middle of the longest fixed span in the bridge and the bridge held this weight successfully without breaking. Also, the deflection resulting from increasing the mass of the object on the bridge from 1Kg to 5Kg was measured by the "lab quest" device which is connected to the "motion detector 2" device that detects the position of an object related to time. After analyzing the data, the deflection didn't exceed the 5cm limit in the design requirements as the maximum deflection was 1.8 mm, as shown in Figure 7 and Table 2. The final positive result was when the bridge was weighed by using the digital scale to check that its mass didn't exceed 20 kg. Fortunately, the weight of the bridge was about 3 kg.

Table 2: The deflection related to the mass of an object

Mass	Deflection (based on the trendline)
0.5kg	0.23mm ± 0.1mm
1kg	0.46mm ± 0.1mm
1.5kg	0.49mm ± 0.1mm
2kg	0.62mm ± 0.1mm
2.5kg	0.76mm ± 0.1mm
3kg	0.89mm ± 0.1mm
5kg	1.41mm ± 0.1mm

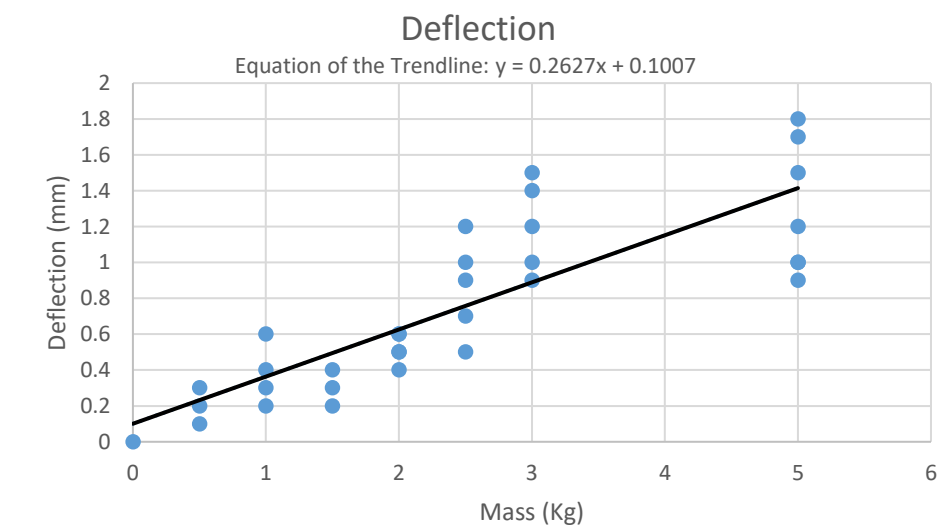


Figure 7: Scatter plot graph showing The deflection of the bridge related to the mass which is put on it



ANALYSIS

Torque:

Torque is the force measurement that causes something to rotate around a point. As shown in equation 1, (τ) is the torque, (F) is the force applied to rotate something, (r) is the force arm of the object (the distance between the axis of rotation and the point at which the force falls on it). The measuring unit of the torque is Newton multiplying by Meter (N.M) and its dimensional formula is ML²T⁻². This concept could be applied to the bridge. To get the torque of the rotation of the bridge, angle θ in the bridge is the angle between the force arm and the line of action, which is equal to 90 degrees, so Sin(θ) is equal to 1. F is the force applied on the bridge to rotate it.

$$\tau = F \times r \times \sin \theta$$

Truss analysis:

Truss analysis means determining the internal forces of the truss members when they are exposed to force on the joints, which helps in designing efficient and safe structures.

Figure 8 shows a model of the truss of the fixed span with its analysis. The blue members are in tension while the red members are in compression. There are two ways to measure these forces: the joint and section methods. The joint method is made by drawing a free-body diagram (FBD) for each joint alone showing the forces that apply to it and resolving these forces on the X and Y axes of the point using SIN and COS trigonometric ratios.

If the force of the member isn't on the X or Y axes, you can resolve it on the X axis by multiplying it with COS(θ) and on the Y axis by multiplying it with SIN(θ), where θ is the angle between the force and X axis. Knowing that the truss should be in static equilibrium, as shown in equation 2. So the summation of the forces on the X-axis and summation of forces on the Y-axis are all equal to 0. One of the advantages of the Pratt truss is that the diagonals, which are the longest members, are in tension, not in compression, like the Howe truss. And this is the reason for choosing this type of truss.

Gear ratio:

As shown in equation 3, gear ratio (u) is the ratio between the number of teeth of the driven gear (T₂) and the number of teeth of the driver gear (T₁). In any gear system, the driver gear is exposed to the input torque, while the driven gear is the gear with the output torque. This ratio helps identify the resultant torque and rpm transformed from the driver gear to the driven gear. As shown in equation 4, The output torque of the driven gear (T_{out}) is calculated by multiplying the input torque of the driver gear (T_{in}) with the gear ratio (u). If the gear ratio exceeds 1, the output torque exceeds the input torque, which is good since it saves effort. But, if the gear ratio is less than 1, the output torque is less than the input torque. As shown in Figure 9, In the prototype, to get a higher output torque, the number of teeth in the driver gear is 18, while in the driven gear, it is 42. So, the gear ratio equals 42 ÷ 18 = 2.3, which means that for every 2.3 rotations made in the driver gear, the driven gear rotates one rotation. As a result, the gear system is a bit slow which ensures stability, on the other hand, the output torque equals the input torque multiplied by 2.3.

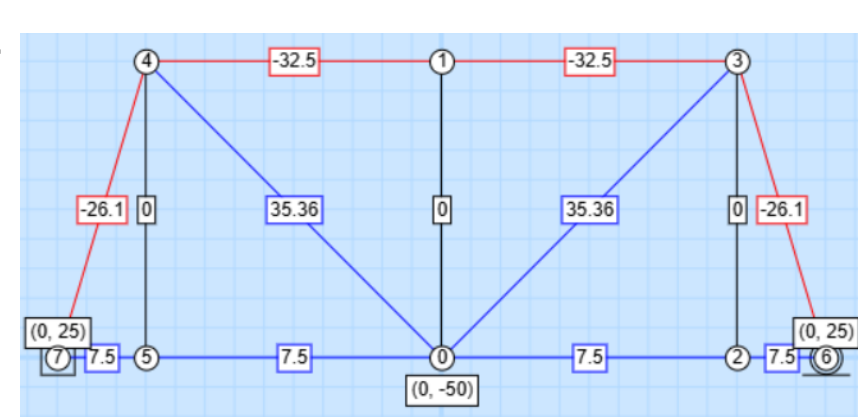


Figure 8: truss analysis of the fixed span's truss

$$\Sigma F_x = 0 \quad \Sigma F_y = 0$$

Equation 2: equation of equilibrium

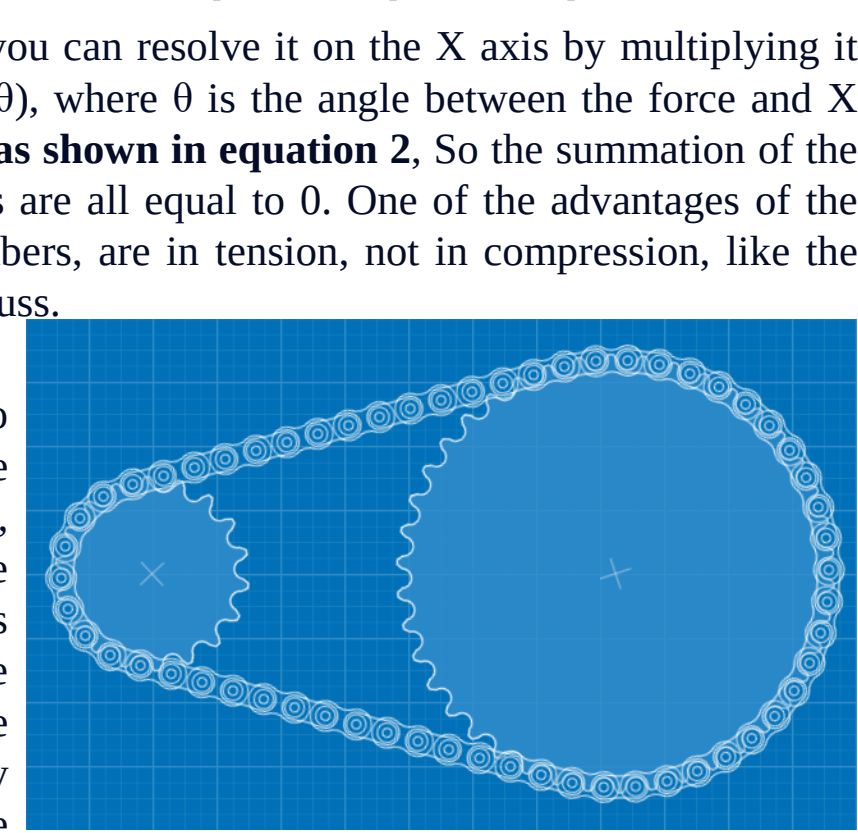


Figure 9: The gear system in the prototype

$$u = \frac{T_2}{T_1}$$

Equation 3: Gear ratio

$$T_{out} = T_{in} \times u$$

Equation 4: In and out Torque in the gear system

Stress and Strain:

Stress (Pascal: Pa: N/m²) is the external force that acts on an object and causes deformation (F) per unit of the cross-sectional area (A²). Strain (dimensionless unit) is the measure of the degree of deformation. Specifically, it is the ratio between the changing of the length of an object to the original length (length without stress).

There are two types of stress and strain, normal and shear, the difference between them is that in normal type, the forces act perpendicular to the cross-sectional area but in shear type, the forces act parallel to the cross-sectional area. As shown in equation 5, σ is stress, F is the force that acts on the object, and A is the cross-sectional area. As shown in equation 6, ε is strain, ΔL is the change in length, and L₀ is the original length. The type of stress and strain in the members of the truss is normal because the applied force is perpendicular to the cross-section.

$$\sigma = \frac{F}{A}$$

Equation 5: Normal stress

$$\epsilon = \frac{\Delta L}{L_0}$$

Equation 6: Normal strain

Stress-strain diagram and UTS:

Stress-strain diagram is used to study the behavior of the materials from the point where it is loaded until they break. As shown in Figure 10, if the material is unloaded between 1 and 2 it will return to its original shape, like a rubber band. This type of behavior is called elastic and the region between points 1 and 2 is called elastic region. The point 2 is called yield strength (σ_y). If the point is passed, the material will not be able to return to its original shape and there will be some deformation in it. The area beyond point 2 is the plastic region. Tensile strength (TS) or Ultimate Tensile strength is the largest value of stress found on the diagram. This is the most stress that the material can endure without breaking and it is different among the materials. As shown in equation 7, the value of tensile strength can be calculated by the calculation of the ratio between the maximum force that is applied on a material and cross-sectional area, and it is measured by Pascal (Pa). The tensile strength of the palm midrib that is used in the prototype is 248 MPa. The member that has the largest applied force has a cross-sectional area of 0.000025 m² and the force that is applied on it is 35.36 N then the stress of the member is 35.36/0.0025 = 1,414,400 Pa. This stress is tiny compared to the tensile strength of the midrib. These data show that palm midrib is a strong material for building the prototype.

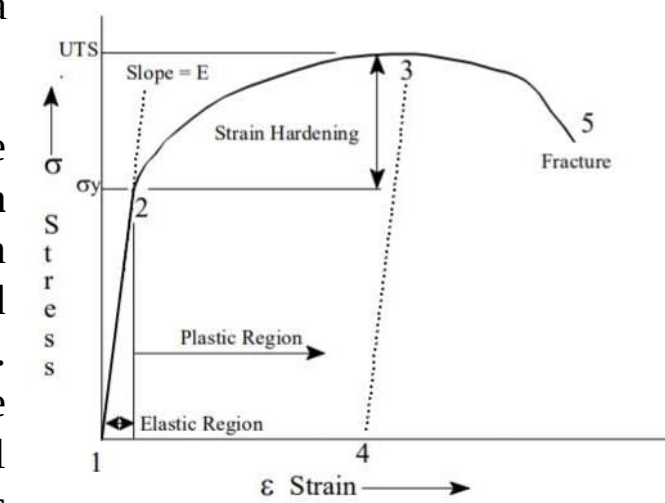


Figure 10: Stress and strain diagram

$$p = \frac{F_{max}}{A}$$

Equation 7: Tensile strength

As shown in Table 3, Some Learning outcomes were used to facilitate the making and adjustment of the solution.

Table 3: Learning outcomes that helped in making the solution

Subject	LO	Concept	Usage
Physics	LO.1.02	Free body diagram	Drawing a free-body diagram of the truss's joints to know the distribution of forces on it.
Math	LO.1.01	Six trigonometric functions	Knowing the relation between the sides and angles in the truss and getting the unknown values helps in the force analysis.
Math	LO.1.02	Scatter plot	Presenting the relation between the mass of the object on the deck and its deflection on a scatter plot graph.
Earth Science	LO.1.03	Earth's materials for building and construction	Using the suitable Earth's materials for building and constructing the bridge.
Physics	LO.1.05	Torque	Knowing the definition of Torque and how it contributes to the selected solution



CONCLUSIONS

In conclusion, the project is designed to solve Egypt's grand challenge which is urban congestion and population growth. The chosen problem to be solved is urban congestion. It met the design requirements and achieved the required test plan successfully as it lifted the 5kg mass, its mass is approximately 3kg as it hasn't reached 20kg mass, and it made a deflection equal to 1.8mm. And lastly, its mechanism of rotation as a swing bridge worked successfully. It is constructed perfectly with fewer errors as it is built according to the following scientific bases which are the truss analysis, the torque, and the gear ratio which made the construction of the bridge go more perfectly and work more effectively. The bridge solved the prior solutions problems from each of them and the result is a complete bridge with the solutions of the past prior solutions Taking into account that we took those advantages and tried to solve those disadvantages to use them in the bridge. The bridge solves the urban congestion in the urban governorates in Egypt and makes transportation quicker.



RECOMMENDATION

Real life application.

It is recommended that one of the best locations to build the bridge in is the water stream "Al-Reyah Al-Tawfike" in Al Qalyobia Governorate. As shown in Figure 11, The stream is located between "Kafr El-Shorafa Al-Gharbi village", and "Bahada village". Kafr El-Shorafa is a village that suffers from urban congestion because of its big population of 18200 people. The village has no hospitals. The village has only one school although the big number of children (nearly 6600 children). The village doesn't have any post offices. Bahada village, located on the other bank of the stream has a hospital, schools, and post office. The displacement between the two villages is nearly 1.5km, but the stream cuts this way, so the actual distance reaches 6km. If the bridge is built in the selected location, the people in Kafr El-Shorafa will be able to move between the two villages and go to hospitals and other services easily and quickly. The deck of the bridge is made up of reinforced concrete, the bridge's truss is made up of austenitic stainless steel, and the mechanical part of the bridge is made up of alloy steel. The bridge will be 50m in length (the width of the stream). The bridge will have a traffic lane, a pedestrian lane, and a middle island with 3.5m, 1.5m, and 1m length respectively.

Also, it is recommended to use a different type of concrete called hydrotex concrete for the construction of the piers and abutments, which consists of various components; it is used underwater to help decrease the erosion rate of the concrete columns of the bridge under the water stream surface, which increases the safety of using the bridge and reduce the percentage of being damaged due to the salty of the water which the bridge's columns are put in it. However, it cannot be used because it isn't available in large quantities.



Figure 11: The real-life location.

Energy tiles:

It is recommended to put a special type of tile on the path of walkers. This type of tile is used to generate electricity from walking on it. The tile will help the light in the bridge. This is because it is expected to find a large number of walkers. As shown in Figure 12, the mechanism of the tile is to make a surface that can be stressed by putting a spring under it. The surface tile will be connected to a rack that is in a vertical case. Then, the rack will be connected with a gear that is connected to a dynamo. Then, if the rack moves by stressing on the surface tile, the gear will rotate then the dynamo will rotate and generate electricity. This system is very effective in places that have a large number of walkers like the bridge.

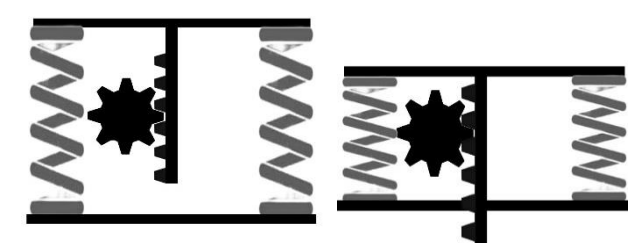


Figure 12: the system of energy tile.

Bridge sensors:

It is recommended to add sensors that are responsible for making an alert when they find that there is a ship that is more than the maximum height which is 5m in real life. "Active detectors" or "Radar-based motion sensors" can be used as they emit radio waves/ microwaves across the place, which strike on nearby objects and reflect it to the sensor detector which determines that there is a moving object here as shown in figure 13. It can be used to make alerts on the bridge to emphasize that there is a ship that will move right now, and it detects if there is a ship moving with a height of more than 5m to open the mechanism which prevents any accidents occur. It also can use the "active detectors" to complete this process successfully

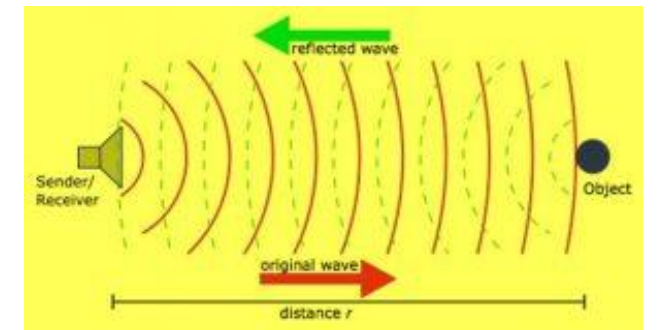


Figure 13: Active Motion Sensor Operation



LITERATURE CITED

- Serway, R. A., & Vuille, C. (2011). Rotational Equilibrium and Rotational Dynamics. *College Physics* (9th ed., pp. 235-239). Cengage Learning.
- Poh'sié, G. H. (2024). *CH10_Deformation*. Scribd. <https://www.scribd.com/document/484879964/Ch10-Deformation>
- P, P., & J, E. R. D. (2015). Characterization of chemical and physical properties of palm fibers. *Advances in Materials Science and Engineering an International Journal* (MSEJ), 2(4), 01–06. <https://doi.org/10.5121/msej.2015.2401>
- Egypt: Administrative Division (Governorates and Districts) - Population statistics, Charts and map. (2024, February 26). Citypopulation. <https://www.citypopulation.de/en/egypt/admin/>
- Engineering at Alberta Courses » Analysis of trusses. (2020, May). Engcourses-uofa. <https://engcourses-uofa.ca/books/statics/structural-analysis/analysis-of-trusses/>
- Terre Armée. (2024, February 9). *Hydrotex® formed concrete matresse solutions*. Terre Armée France. <https://www.terre-armee.fr/en/reinforced-earth/hydrotex-formed-concrete-mattresses/>



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